Computer Graphics

Graphics Primitives & Transformations

Points

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Point Representation

- A point is represented using 2 or 3 numbers (x, y, [z]) that are the projections on to the respective coordinate axes
- Fundamental shape-defining primitive in most Graphics APIs. Everything else is built from it!
- Represented using byte, short, int, float, double, etc.
- The scale and unit are application dependent. Could be angstroms or lightyears!
- Point can undergo transformations: Translations and Rotations

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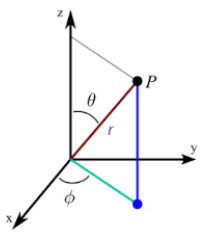
Point Representation

Cartesian

$$(x, y, z)$$
$$z = r\cos\theta$$
$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$



Polar

$$(r, \theta, \phi)$$

r :Distance from center

θ :Elevation

ϕ :Azimuthal

$$r^2 = x^2 + y^2 + z^2$$
$$\phi = \tan^{-1}(y/x)$$
$$\theta = \tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}$$

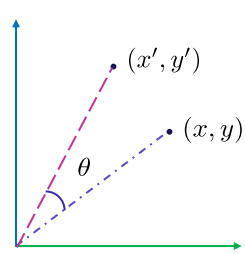
Geometric Transformations on Points

Translation

- Translate a point $P = (x, y, [z])$ by $\mathrm{T} = (a, b, [c])$
- The translated point coordinate becomes
  - $P' = (x + a, y + b, [z + c])$
  - In Vector form, $P' = P + \mathrm{T}$
- Distances, angles between set of points undergoing same transformation is preserved

2D Rotation

- Rotate a point $P = (x, y)$ about origin CCW by θ
- Rotated point $P' = (x', y')$
  - $x' = \mathrm{x}\cos\theta - y\sin\theta$
  - $y' = \mathrm{x}\sin\theta + y\cos\theta$
  - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
  - $P' = RP$



Properties of Rotation Matrix

- Preserves Euclidean geometry:
  - Distances between any pair of points are unchanged
  - Areas in 2D and volumes in 3D are preserved
  - Angles between any two vectors x and y is unchanged after rotation
  - Circles remain circles, spheres remain spheres; general shapes are just rotated, not distorted
  - Maps the origin to itself and lines to lines; parallelism is preserved
- Columns & rows form an orthonormal basis:
  - Has unit length
  - Are mutually orthogonal
- This is equivalently written
  - $R^T R = R R^T = I$
  - Inverse equals transpose: $R^{-1} = R^T$
  - Determinant is +1 for a proper rotation (no reflection)
  - If a matrix is orthonormal but has $\det = -1$, it represents a reflection (or improper rotation), not a pure rotation

3D Rotation

Rotation about X-axis

$$R_x(\rho)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\rho & -\sin\rho \\ 0 & \sin\rho & \cos\rho \end{bmatrix}$$

Rotation about Y-axis

$$R_y(\phi)$$

$$\begin{bmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{bmatrix}$$

Rotation about Z-axis

$$R_z(\theta)$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Combination of Transformations

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Translation & Rotation

- Given a Point $P = (x, y, z)$, translate it by $\mathrm{T} = (a, b, c)$ and then rotate across y-axis by angle $\emptyset$
- Translate
  - $P' = (x + a, y + b, [z + c])$
  - $P' = P + T$

Vector-Vector addition

- Then rotate P' by $\emptyset$
  - $P'' = R_y(\emptyset)P'$
  - $P'' = R_y(\emptyset)(P + T)$

Matrix-vector Product

- Requires matrix addition as well as multiplication

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Is there a better way to compute the combinational transform

Homogeneous Coordinates

- Homogeneous coordinates extend ordinary (Cartesian) coordinates by one extra non-zero scale factor w
  - A 2D cartesian point (x, y) is represented by homogeneous form as (x_h, y_h, w) with w $\neq 0$
  - A 3D cartesian point (x, y, z) is represented by homogeneous form as (x_h, y_h, z_h, w) with w $\neq 0$
- The actual Cartesian coordinates are recovered by dividing by (w)
  - $(x, y) = \left(\frac{x_h}{w}, \frac{y_h}{w}\right)$
  - $(x, y, z) = \left(\frac{x_h}{w}, \frac{y_h}{w}, \frac{z_h}{w}\right)$
- Any nonzero scalar multiple represents the same point, so $(x, y, z, 1)$ and $(2x, 2y, 2z, 2)$ are equivalent

Translation in Homogeneous Coordinates

Cartesian Coordinates

- $P = (x, y)$ and $T = (a, b)$
- $P' = P + T$
- Not a pure matrix operation

Homogeneous Coordinates

- $P = [x, y, 1]^{\mathsf{T}}$
- $T = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$
- $P' = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = TP$
- Matrix multiplication operation

3D Rotation in Homogeneous Coordinates

Rotation about X-axis

$R_x(\rho)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\rho & -\sin\rho \\ 0 & \sin\rho & \cos\rho \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\rho & -\sin\rho & 0 \\ 0 & \sin\rho & \cos\rho & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about Y-axis

$R_y(\phi)$

$$\begin{bmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{bmatrix}$$

$$\begin{bmatrix} \cos\phi & 0 & -\sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ \sin\phi & 0 & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about Z-axis

$R_z(\theta)$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation about a Point

- Rotate a point $P = (x, y, z)$ about $C = (a, b, c)$ CCW by θ around y-axis
- Translate P so that C becomes the origin: $T_{-C}P$
- Rotate around the origin: $R_\phi T_{-C}P$
- Translate back to C: $T_C R_\phi T_{-C}P$
- $P' = T_C R_\phi T_{-C}P$

$$T_{-C} = \begin{bmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 1 & -c_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_\phi = \begin{bmatrix} \cos\phi & 0 & -\sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ \sin\phi & 0 & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_C = \begin{bmatrix} 1 & 0 & 0 & c_x \\ 0 & 1 & 0 & c_y \\ 0 & 0 & 1 & c_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Properties of Homogeneous Coordinates

- A point is transformed once: $P' = MP$,which is efficient and numerically stable for graphics pipelines
- Homogeneous vectors with last coordinate $w \neq 0$ represent points; they can be normalized to Cartesian form
- Vectors with $w = 0$ represent directions (no fixed position) and are not affected by translation (only by rotation/scale)

Thank You